(b) Values of n and the two measured values of the maximum potential difference V_1 and V_2 are given in Table 2.1.

Table 2.1

n	V_1 / V	V_2 / V	V / V	$\frac{1}{V}$ / V^{-1}
2	4.30	4.20		
3	3.65	3.75		
4	3.30	3.20		
5	2.85	2.95		
6	2.65	2.55		
7	2.30	2.40		

Calculate and record values of V / V and $\frac{1}{V}$ / V^{-1} in Table 2.1. Include the absolute uncertainties in V and $\frac{1}{V}$. [2]

(c) **(i)** Plot a graph of $\frac{1}{V}$ / V^{-1} against n. Include error bars for $\frac{1}{V}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ... [2]